

Question	Answer	Marks
7(a)	direction of induced e.m.f.	M1
	such as to (produce effects that) oppose the change that caused it	A1
7(b)(i)	$\Phi = BA$	C1
	$= 0.047 \times 10^{-3} \times \pi \times 12^2$	C1
	$= 0.021 \text{ Wb}$	A1
7(b)(ii)	e.m.f. = flux cut / time	C1
	$= 0.021 / (1 / 85)$	A1
	$= 1.8 \text{ V}$	
7(b)(iii)	force (due to current) cause anticlockwise moment (to oppose rotation)	B1
	(from the LH rule) current is from O to X, so X is at the higher potential	B1

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